Practice Workbook §10.1 Student's t Distribution

Question 1 (§10.1)

The statistic $t = (\bar{x} - \mu) / (s / \sqrt{n})$ follows a t distribution with $df = n - 1$.

What two assumptions do we need for this to hold?

Question 2 (§10.1)

Use the t table to find each value of t_α (right-tail area α).

(a) $\alpha = .05$, $df = 20$ (b) $\alpha = .025$, $df = 12$ (c) $\alpha = .01$, $df = 9$

Question 3 (§10.1)

A t random variable has $df = 15$.

(a) Find t_0 so that $P(t > t_0) = .05$.

(b) Find t_0 so that $P(-t_0 < t < t_0) = .95$.

(c) Compare (b) with $z = 1.96$. Which is larger, and why?

Practice Workbook §10.2 Inference for a Population Mean (small sample)

Question 1 (§10.2)

A barista times 10 lattes. $\bar{x} = 7.5$ min, $s = 0.9$ min. Assume the times are normal.

Find a 95% confidence interval for the mean time μ.

Question 2 (§10.2)

Same data ($n = 10$, $\bar{x} = 7.5$, $s = 0.9$). Test H_0: $\mu = 7$ against H_a: $\mu > 7$ at $\alpha = .05$.

Question 3 (§10.2)

$n = 16$, $\bar{x} = 48.2$, $s = 4$. Test H_0: $\mu = 50$ against H_a: $\mu \neq 50$ at $\alpha = .05$. Give a range for the p-value.

Practice Workbook §10.3 Difference Between Two Means (independent samples)

Question 1 (§10.3)

Sample 1: $n_1 = 8$, $\bar{x}_1 = 12$, $s_1 = 2$. Sample 2: $n_2 = 10$, $\bar{x}_2 = 10$, $s_2 = 2.5$.

(a) Is it fair to pool the variances? (b) Find the pooled variance s^2.

Question 2 (§10.3)

Same data. Test $H_0: \mu_1 = \mu_2$ against $H_a: \mu_1 \neq \mu_2$ at $\alpha = .05$.

Question 3 (§10.3)

Same data. Find a 95% confidence interval for $\mu_1 - \mu_2$. Does it agree with Question 2?

Practice Workbook §11.1 The Design of an Experiment

Question 1 (§11.1)

30 volunteers are randomly assigned to three diets (10 each). Weight loss after 8 weeks is recorded.

Name the experimental unit, the response, the factor, its levels, and the treatments.

Question 2 (§11.1)

A café tests milk type (whole, oat) and syrup (none, vanilla, caramel).

(a) How many factors? How many treatments? (b) List the treatments.

Question 3 (§11.1)

Experiment or observational study? Explain.

(a) We compare the incomes of customers who chose oat milk and those who chose whole milk.

(b) We randomly give each of 40 customers oat or whole milk and record a taste score.

Practice Workbook §11.2 One-Way Analysis of Variance

Question 1 (§11.2)

k = 4 treatments, n = 20 in total. SST = 30, SSE = 64. Complete the ANOVA table.

Question 2 (§11.2)

k = 3 treatments, 5 observations each. SST = 24, Total SS = 60.

Test H_0: $\mu_1 = \mu_2 = \mu_3$ at $\alpha = .05$.

Question 3 (§11.2)

Same design (MSE = 3, df = 12, $n_i = 5$). $\bar{x}_1 = 20$, $\bar{x}_2 = 17$.

(a) 95% CI for μ_1. (b) 95% CI for $\mu_1 - \mu_2$.